Supplementary material: Integrated Flood Forecasting and Mapping with Deep Learning

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A The CAMELS datasets

The CAMELS (Catchment Attributes and MEteorology for Large-sample Studies) family provides harmonized, open datasets that couple basin-scale hydrometeorological time series with richly curated catchment attributes across many countries, enabling reproducible, cross-basin hydrologic research and machine learning at scale [1]. National CAMELS variants (e.g., USA, GB, AU, BR, CL, DE) follow a common philosophy: consistent basin delineations; daily time series of streamflow and meteorological forcings; and static descriptors summarizing climate, topography, soils, geology, land cover, and hydrologic regime. By standardizing variable names, units, and metadata, CAMELS facilitates model transfer and fair benchmarking across physioclimatic gradients.

Data are typically organized by basin identifiers, with (i) daily discharge at the gauging outlet; (ii) basin-aggregated meteorological forcings (e.g., precipitation, temperature, radiation, potential evapotranspiration) derived from vetted gridded products or station networks; and (iii) static attributes compiled from authoritative sources. Attributes commonly include basin area and hypsometry; aridity and seasonality indices; baseflow and runoff coefficients; soil texture and hydraulic properties; geology classes; land cover fractions and anthropogenic indicators; and hydroclimatic statistics (e.g., flow percentiles). Most releases document quality control, gap coding, coordinate reference, versioning, and known caveats (e.g., rating-curve changes, regulated flows).

While the core structure is shared, sources and coverage vary by country (period length, number of basins, forcing products, availability of regulation metadata). The CARAVAN initiative [3] further consolidates multiple CAMELS-style datasets under a single access point with harmonized schemas, easing multi-regional studies and transfer learning. Users should consult each dataset's documentation for precise variable definitions, units, processing lineage, and known limitations.

B Performance metrics

Let o_t and s_t denote observed and simulated discharge at times $t = 1, \ldots, T$. Denote sample means μ_o,μ_s, standard deviations σ_o, σ_s, and the observed mean $\bar{o} = \frac{1}{T}\sum_{t=1}^{T} o_t$. For flow-duration-curve (FDC) metrics, let π be the permutation that sorts observations in descending order, and define $\tilde{o}i = o\pi(i)$ and $\tilde{s}i = s\pi(i)$.

Mean Squared Error (MSE).

$$\mathrm{MSE} \;=\; \frac{1}{T}\sum_{t=1}^{T}(s_t - o_t)^2, \tag{1}$$

the average squared deviation between simulations and observations (lower is better).

Nash–Sutcliffe Efficiency (NSE).

$$\text{NSE} \;=\; 1;-;\frac{\sum_{t=1}^{T}(s_t - o_t)^2}{\sum_{t=1}^{T}(o_t - \bar{o})^2}. \tag{2}$$

NSE compares model errors to the variance of observations. Values range from $-\infty$ to 1; 1 is perfect, 0 equals the mean-as-forecast baseline, and negative values are worse than that baseline. Because it uses squared errors, NSE is more sensitive to high flows.

Pearson correlation (r).

$$r \;=\; \frac{\sum_{t=1}^{T}(s_t - \mu_s)(o_t - \mu_o)}{\sqrt{\sum_{t=1}^{T}(s_t - \mu_s)^2},\sqrt{\sum_{t=1}^{T}(o_t - \mu_o)^2}}, \tag{3}$$

measures linear association (timing/shape coherence) independent of bias and scale; $r \in [-1, 1]$.

Kling–Gupta Efficiency (KGE) and components. Define the bias and variability ratios

$$\beta \;=\; \frac{\mu_s}{\mu_o}, \qquad \alpha \;=\; \frac{\sigma_s}{\sigma_o}. \tag{4}$$

Then

$$\text{KGE} \;=\; 1;-;\sqrt{(r-1)^2 + (\alpha-1)^2 + (\beta-1)^2}. \tag{5}$$

KGE synthesizes correlation (r), variability (α), and bias (β); it equals 1 when $r = \alpha = \beta = 1$.

α-NSE and β-NSE. To diagnose variance and bias effects with efficiency-style scores, we report

$$\alpha\text{-NSE} \;=\; 1 - (\alpha - 1)^2, \qquad \beta\text{-NSE} \;=\; 1 - (\beta - 1)^2, \tag{6}$$

which both attain 1 at perfect variability/bias ($\alpha = \beta = 1$) and decrease quadratically with departure from unity.

High-Flow Volume (FHV). Let $n_H = \lfloor p_H T \rfloor$ denote the number of high-flow samples in the FDC tail (e.g., $p_H = 0.02$). Sorting by observed flows (descending), define

$$\text{FHV} \;=\; 100 \times \frac{\sum_{i=1}^{n_H}\left(\tilde{s}_i - \tilde{o}i\right)}{\sum i = 1^{n_H}\tilde{o}_i} \tag{7}$$

Positive values indicate overestimation of flood volumes; negative values indicate underestimation.

Low-Flow Volume (FLV). Let $n_L = \lfloor p_L T \rfloor$ be the number of low-flow samples in the FDC tail (e.g., $p_L = 0.30$). Using the same sorting (descending), the lowest n_L points are $i = T - n_L + 1, \ldots, T$:

$$\mathrm{FLV} \;=\; 100 \times \frac{\sum_{i=T-n_L+1}^{T} \left(\tilde{s}_i - \tilde{o}i\right)}{\sum i = T - n_L + 1^T \tilde{o}_i} \tag{8}$$

Positive values indicate overestimation of low-flow volumes; negative values indicate underestimation.

Flow-Magnitude Slope (FMS). FMS compares the slope of the mid-segment of the FDC. For user-chosen mid-quantiles $p_u < p_v$ (e.g., $p_u = 0.2$, $p_v = 0.7$), let $Q_x(p)$ be the p-quantile of series x computed on the FDC. Using log flows to emphasize proportional differences,

$$m_o \;=\; \log Q_o(p_u) - \log Q_o(p_v), \tag{9}$$

$$m_s \;=\; \log Q_s(p_u) - \log Q_s(p_v), \tag{10}$$

$$\mathrm{FMS} \;=\; 100 \times \frac{m_s - m_o}{m_o} \tag{11}$$

Positive values indicate a steeper simulated mid-segment (excess variability); negative values indicate a flatter one.

Peak-Timing. Peak-timing error quantifies the temporal offset between simulated and observed flood peaks. Peaks are first identified in the observed series using a standard prominence-and-separation rule. For each observed peak i at time $t_i^{(o)}$, we locate the time $t_i^{(s)}$ of the simulated local maximum within a symmetric matching window $[t_i^{(o)} - w, , t_i^{(o)} + w]$ (unmatched peaks are discarded). We then summarize timing error as the median absolute lag

$$\mathrm{PeakTiming} \;=\; \mathrm{median}_i(|t_i^{(s)} - t_i^{(o)}|) \tag{12}$$

reported in the native time unit (e.g., hours or days); smaller values indicate better synchronization.

C Reproducibility and implementation notes

All code for runoff forecasting and inundation modeling is available in the accompanying repository at Github. Dependencies and reproduction instructions are included. Forecasting experiments use the publicly released CAMELS-DE data.

All rainfall–runoff models are trained with fixed random seeds, identical splits, and identical preprocessing. Validation-driven hyperparameters and trained weights are frozen prior to testing. Coding is in Python using `NeuralForecast` for sequence models; statistical testing uses standard scientific Python libraries. Inundation modules are implemented as lightweight routines operating on raster arrays (no hydraulic solver). Figure generation scripts apply the same normalization and color scales across horizons where applicable.

D Full statistical comparisons for Exp. 2

In the second experiment, we compare LSTM, TCN, and PatchTST across loss formulations (MQ_L vs MSE_L) 1–3 day horizons, see Table **??**. For NSE at day 1, PatchTST with MSE_L is strongest: it significantly outperforms LSTM with MQ_L ($d \approx -0.73$, $p = 5.23 \times 10^{-5}$) and TCN for both losses (e.g., versus TCN MSE_L $d \approx -2.70$, $p < 10^{-13}$). Within-model contrasts are informative: LSTM favors MSE_L on day 1 (moderate effect, $d \approx -0.44$, $p = 8.47 \times 10^{-4}$), but on days 2 and 3 the advantage shifts to MQ_L (day 2: $d \approx 0.52$, $p = 2.86 \times 10^{-4}$). TCN also exhibits a strong internal loss effect (e.g., day 2 NSE, $d \approx 1.97$, $p < 10^{-13}$), and inter-model comparisons frequently show PatchTST outperforming both competitors (see Fig. **??**).

For KGE at day 1, PatchTST with MSE_L outperforms LSTM (paired $d \approx -1.29$, $p < 10^{-12}$) and TCN, while PatchTST's own loss choice causes large swings across horizons (e.g., day 3 MQ_L vs MSE_L, $d \approx -3.39$, $p < 10^{-14}$). For longer lead times, LSTM and PatchTST occasionally exchange relative advantage (e.g., day 3 LSTM vs PatchTST with MQ_L, $d \approx 2.62$, $p < 10^{-13}$), yet most pairwise differences remain significant with large effect sizes (see Fig. **??** and Appendix Figs. 1–2). For α-NSE, PatchTST's internal loss effect is especially pronounced, MSE_L yields far higher skill than MQ_L (e.g., day 1 paired $d \approx -3.77$, $p < 10^{-13}$; day 3 $d \approx -4.88$, $p < 10^{-14}$). LSTM with MQ_L at intermediate horitzons is the best. In β-NSE, loss choice again drives large effects (e.g., PatchTST day 2 and 3 paired $d - 4.2$, $p < 10^{-14}$), with MSE_L improving bias behavior relative to MQ_L. LSTM's internal contrast is significant across horizons (e.g., day 1 $d \approx 1.07$, $p < 10^{-8}$), and it substantially outperforms TCN in bias-related skill (e.g., day 1 vs TCN MQ_L $d \approx 2.68$, $p < 10^{-13}$). PatchTST frequently surpasses TCN and, depending on configuration, can also beat LSTM in β-NSE. A representative hydrograph for basin DE210310 (Fig. **??**) illustrates temporal alignment and uncertainty (PatchTST, 90% CI).

For FHV, PatchTST shows a dominant internal benefit from MSE_L across all horizons (e.g., day 1 $d \approx -3.93$, $p < 10^{-13}$; day 3 $d \approx -5.29$, $p < 10^{-14}$), indicating markedly better representation of high flows. LSTM's loss effect evolves with lead time: it is non-significant at day 1 but MSE_L underperforms MQ_L by days 2 and 3 (e.g., day 2 $d \approx 1.72$, $p < 10^{-14}$). PatchTST with MSE_L also outperforms both LSTM and TCN in inter-model comparisons (e.g., versus TCN day 1, $d \approx -2.77$, $p < 10^{-13}$). For FLV, which captures low-flow behavior, significant internal improvements under MQ_L appear for all models at early horizons (e.g., day 2 PatchTST $d \approx 0.74$, $p < 10^{-9}$; TCN $d \approx 0.54$, $p < 10^{-9}$).

In FMS, LSTM under MQ_L excels at short horizons (e.g., day 1 vs PatchTST MQ_L, $d \approx 1.12$, $p < 10^{-8}$), and its internal loss contrast is meaningful (day 1 MQ_L better than MSE_L, $d \approx 0.43$, $p = 5.53 \times 10^{-3}$). PatchTST and TCN also show substantial internal differences, and by longer horizons PatchTST with MSE_L can outperform TCN with MSE_L (e.g., day 3, $d \approx -2.38$, $p < 10^{-12}$). Peak-Timing reveals that LSTM's loss choice has a large effect on synchronization (day 1 MQ_L vs MSE_L, $d \approx 1.18$, $p = 1.44 \times 10^{-7}$), and it frequently outperforms the other architectures at short lead times. PatchTST shows some advantage over TCN in timing early on (day 1, $d \approx 0.41$, $p = 1.59 \times 10^{-2}$), while differences generally attenuate with horizon except isolated contrasts (e.g., LSTM vs TCN day 2 MQ_L, $d \approx 0.37$, $p = 3.80 \times 10^{-2}$).

E Supplementary figures

This appendix consolidates supplementary figures that support the interpretation of main-text results and clarify the inundation setup. Figures 1 and 2 extend the distributional evaluation of model skill by showing empirical CDFs of NSE and KGE for day-2 and day-3 forecasts across the 50 CAMELS-DE basins, complementing the day-1 curves in the main text. These plots reveal how architectural and loss-function differences evolve with lead time, highlighting convergence or divergence in performance distributions beyond point estimates.

Figures 3 and 4 document the inundation experiment assets. The training masks capture the stage-dependent wet/dry patterns used to learn per-pixel thresholds and to construct water-surface templates, with mild label noise to mimic observational uncertainties. The synthetic DEM provides a controlled topographic context with a gentle slope and a central depression, enabling transparent assessment of extent and depth behavior. Figure 5 illustrates Manifold's extrapolation at a stage well beyond training, demonstrating stable behavior that preserves drainage structure and deepening in the depression. All figures use consistent axes, units, and color scales to facilitate comparison across horizons and stages.

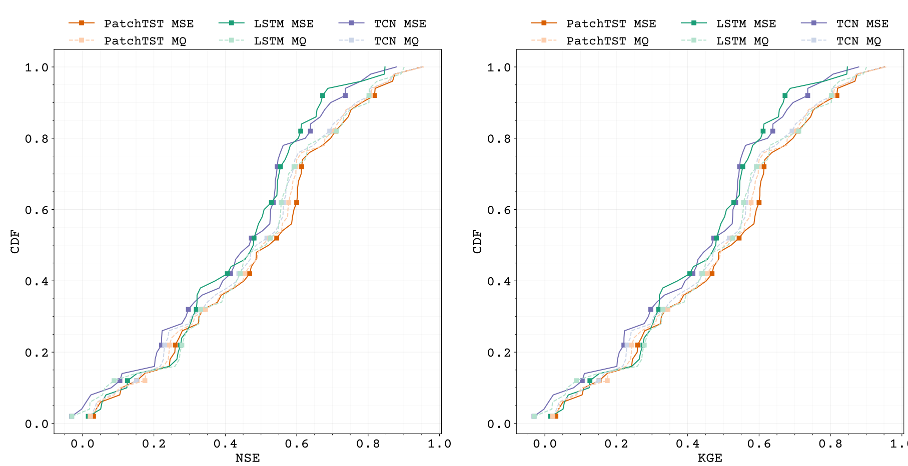


Fig. 1. Empirical cumulative distribution functions (CDFs) of the Nash–Sutcliffe Efficiency (NSE) and Kling-Gupta Efficiency (KGE) for Day 2 predictions across the 50 CAMELS-DE catchments, comparing deep learning models (PatchTST, LSTM, and TCN) trained with standard MSE loss versus multi-quantile (MQ) loss.

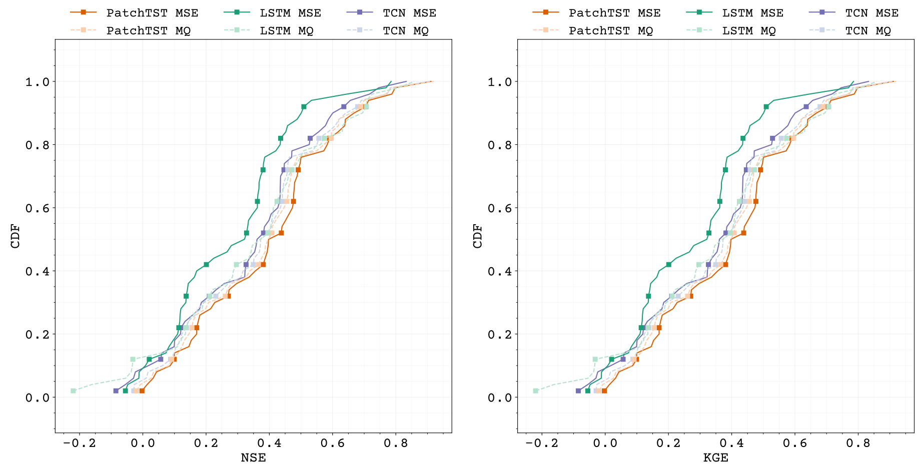
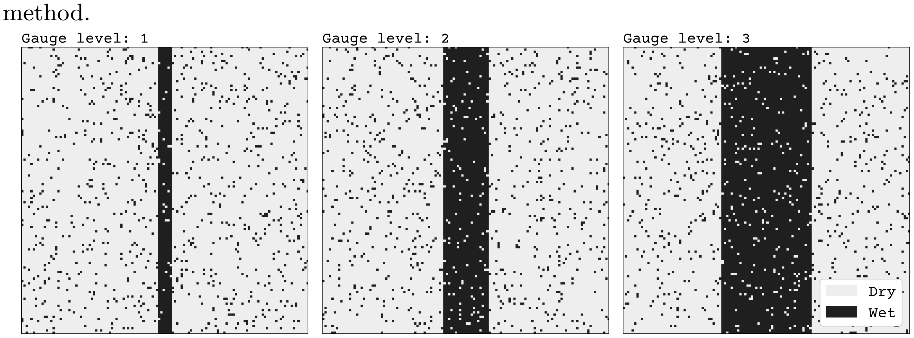


Fig. 2. Empirical cumulative distribution functions (CDFs) of the Nash–Sutcliffe Efficiency (NSE) and Kling-Gupta Efficiency (KGE) for Day 2 predictions across the 50 CAMELS-DE catchments, comparing deep learning models (PatchTST, LSTM, and TCN) trained with standard MSE loss versus multi-quantile (MQ) loss.

Fig. 3. Training dataset for the inundation models: binary wet/dry masks derived from synthetic river channel widths at three gauge levels (1 m, 2 m, and 3 m). The wet pixels represent historical inundation extents used to learn pixel-wise flood thresholds in the Thresholding approach and to build water-surface height contours for the Manifold method.



F Supplementary tables

The following two extra tables provide reference detail and aggregate statistics to complement the main results. Table 1 enumerates the dynamic forcings and static catchment attributes used in the experiments, including names, units, and brief descriptions, enabling reproducibility and facilitating transfer to other regions or datasets. Table 2 reports mean skill across basins by horizon, architecture,

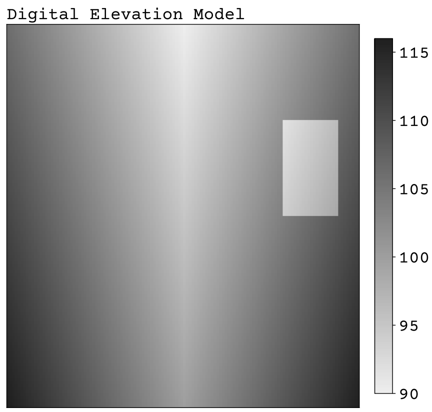


Fig. 4. Synthetic digital elevation model (DEM) used to drive the Manifold inundation example. The DEM is constructed by combining a symmetric x-direction gradient (offset by +100 m), a decreasing y-direction gradient (10 m to 0 m), and a localized −10 m depression in the central subregion. Elevation values range from approximately 90 m to 116 m.

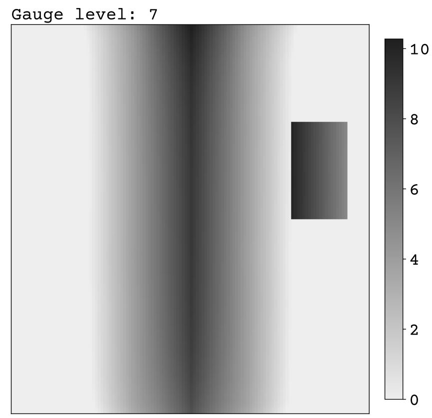


Fig. 5. Extrapolated Manifold model output for a forecasted gauge level of 7.0 m, well beyond the training range. The model raises the highest precomputed water-surface elevation uniformly by the stage exceedance and then computes depths relative to the DEM, correctly capturing the enhanced inundation in the central depression.

and loss, mirroring the median-based summary in the main text. Presenting both mean and median aggregates helps assess robustness to outliers and tail behavior, and supports the significance statements reported in the Results section.

Table 1. Description of the target, static, and dynamic input attributes used in the rainfall–runoff modeling experiments. Static attributes are catchment-specific and time-invariant; dynamic inputs vary over time.

	Attributes	**Description**	**Unit**
Target	`discharge_spec`	Observed catchment-specific discharge	mmd^{-1}
Static input	`area`	catchment area derived from the MERIT Hydro catchment	km^2
	`elev_mean`	mean elevation in the catchment based on the MERIT Hydro geometry	m.a.s.l.
	`clay_0_30cm_mean`	weight percent of clay particles (<0:002 mm) in the fine earth fraction at depths of 0–30, 30– 100, and 100–200 cm	wt%
	`sand_0_30cm_mean`	weight percent of sand particles (>0:05/0.063 mm) at depths of 0–30, 30–100, and 100–200 cm	wt%
	`silt_0_30cm_mean`	weight percent of silt particles ($\geq$ 0.002mm and $\leq$ 0.05/0.063 mm) in the fine earth fraction at depths of 0–30, 30–100, and 100–200 cm	wt%
	`artificial_surfaces_perc`	areal coverage of artificial surfaces	%
	`agricultural_areas_perc`	areal coverage of agricultural areas	%
	`forests_areas_perc`	areal coverage of forests and semi-natural areas	%
	`wetlands_perc`	areal coverage of wetlands	%
	`water_bodies_perc`	areal coverage of water bodies	%
	`p_mean`	long-term mean of daily precipitation from 1951 to 2020	$\mathrm{mm\,d}^{-1}$
	`p_seasonality`	seasonality and timing of precipitation (estimated using sine curves to represent the annual temperature and precipitation cycles, positive (negative) values indicate that precipitation peaks in summer (winter), and values close to zero indicate uniform precipitation throughout the year).	–
	`frac_snow`	fraction of precipitation falling as snow, i.e. while mean air temperature is <0 °C	–
	`high_prec_freq`	frequency of high-precipitation days ($\geq$ 5 times mean daily precipitation)	dyr^{-1}
	`low_prec_freq`	frequency of dry days (<$1\mathrm{mmd}^{-1}$)	dyr^{-1}
	`high_prec_dur`	mean duration of high-precipitation events (number of consecutive days $\geq$ 5 times mean daily precipitation)	d
	`low_prec_dur`	mean duration of dry periods (number of consecutive days <$1\mathrm{mmd}^{-1}$ mean daily precipitation)	d
Dynamic input	`radiation_global_mean`	Observed interpolated spatial mean of the global radiation	Wm^2
	`precipitation_mean` `precipitation_stdev`	Observed interpolated spatial mean and standard deviation of the daily precipitation.	$\mathrm{mm\,d}^{-1}$
	`temperature_min` `temperature_max`	Observed interpolated spatial mean daily minimum and maximum temperatures	ºC

Table 2. Comparison of performance of LSTM, TCN, and PatchTST across multiple hydrological metrics for days 1 to 3. Mean values are reported; ideal values are indicated in the table notes.

Metric	**Time**	**LSTM**		**TCN**		**PatchTST**	
		MQ$_L$	**MSE**$_L$	**MQ**$_L$	**MSE**$_L$	**MQ**$_L$	**MSE**$_L$
NSE[a]	Day 1	0.65	0.67	0.66	0.63	0.69	0.69
	Day 2	0.47	0.44	0.47	0.43	0.48	0.49
	Day 3	0.35	0.28	0.36	0.34	0.38	0.39
MSE[b]	Day 1	1.53	1.43	1.55	1.62	1.47	1.47
	Day 2	2.31	2.22	2.23	2.35	2.19	2.16
	Day 3	2.76	2.64	2.51	2.60	2.46	2.41
Alpha-NSE[c]	Day 1	0.81	0.75	0.70	0.69	0.77	0.84
	Day 2	0.81	0.62	0.61	0.62	0.63	0.71
	Day 3	0.80	0.52	0.54	0.59	0.56	0.65
Beta-NSE[d]	Day 1	0.03	-0.07	-0.11	-0.03	-0.06	-0.01
	Day 2	-0.01	-0.17	-0.16	-0.01	-0.11	0.01
	Day 3	-0.04	-0.25	-0.19	-0.01	-0.15	0.01
Pearson r[e]	Day 1	0.80	0.83	0.82	0.80	0.83	0.83
	Day 2	0.69	0.68	0.69	0.64	0.69	0.68
	Day 3	0.61	0.59	0.61	0.57	0.61	0.61
KGE[f]	Day 1	0.71	0.66	0.62	0.61	0.70	0.76
	Day 2	0.61	0.45	0.46	0.46	0.49	0.57
	Day 3	0.54	0.29	0.36	0.40	0.38	0.47
FHV[g]	Day 1	-25.18	-27.74	-30.35	-29.57	-23.80	-16.85
	Day 2	-26.23	-40.31	-40.11	-36.86	-38.32	-31.04
	Day 3	-27.40	-49.22	-46.82	-39.95	-46.17	-37.90
FLV[h]	Day 1	-404.98	-530.06	-54.38	-239.46	-67.27	-328.18
	Day 2	-498.18	-518.55	-67.60	-282.98	-11.67	-399.94
	Day 3	-535.05	-524.51	-79.26	-309.17	-29.29	-433.99
FMS[i]	Day 1	4.14	-0.83	-6.23	-20.65	-5.30	-2.65
	Day 2	6.79	-5.79	-10.34	-26.39	-9.93	-7.88
	Day 3	10.61	-10.37	-13.43	-26.01	-12.92	-11.06
Peak-Timing[j]	Day 1	1.12	0.97	1.07	1.13	1.04	1.06
	Day 2	1.88	1.76	1.78	1.80	1.81	1.77
	Day 3	2.32	2.29	2.27	2.05	2.36	2.32

[a, f] Nash–Sutcliffe efficiency; Kling-Gupta Efficiency; $(-\infty, 1]$; values closer to one are desirable [4,2].
[b] Mean Squared Error; $[0, \infty)$; values close to zero are desirable.
[c] α-NSE decomposition: $(0, \infty)$; values closer to one are desirable [2].
[d,g,h,i] β-NSE decomposition; Top 2 % peak flow bias; 30 % low flow bias; Bias of FDC mid-segment slope; $(-\infty, \infty)$; values closer to zero are desirable [2,5].
[e] Pearson correlation; $[-1, 1]$; values closer to one are desirable.

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